DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING • FINAL YEAR CAPSTONE PROJECT 2025/2026

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PULSEGRID DEFENSE POSTER | Intelligent Emergency Routing & Bed Management System

An Intelligent, Zero-API-Cost Telemetry & Geospatial Triage Platform Combating Ghana's "No-Bed Syndrome"

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1. THE CRISIS: "NO-BED SYNDROME"

PUBLIC HEALTH CRISIS

In Ghana, traumatic injuries and acute medical emergencies face the severe crisis of **"No-Bed Syndrome"**. Critical ambulances transport unstable patients blindly across metropolitan centers only to be turned away at hospital gates due to 100% ICU/ER saturation. This creates fatal delays during the **Golden Hour** (first 60 mins of trauma).

COMMERCIAL API BARRIER
Google Maps costs \$5-\$10 per 1k requests, costing thousands monthly for national fleets.

NETWORK DEAD ZONES
Highway transit corridors experience cellular dropouts, crashing traditional thin clients.

2. CLINICAL TRIAGE PROTOCOL (SATS / TEWS)

GHANA HEALTH SERVICE STANDARD

PULSEGRID formalizes triage using the **South African Triage Scale (SATS)** and calculates the composite **Triage Early Warning Score (TEWS)**:

TEWS = f(Heart Rate, Systolic BP, Resp Rate, Temp, Mobility, Trauma)

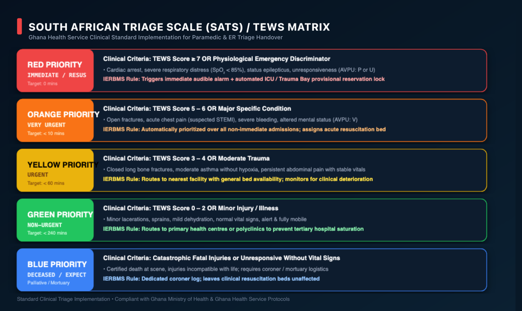

Fig. 1: Clinical Triage Matrix & Multi-Parameter TEWS Scoring Protocol

3. SPATIAL GEOMETRIES (POSTGIS)

2,500+ FACILITIES MAPPED

Utilizes **PostgreSQL 16 + PostGIS** spatial indexing (GIST) to ingest all 2,500+ healthcare facilities across all 16 Ghanaian regions from the verified **UN OCHA / HDX HOT-OSM** census. Fast geodesic spatial queries (ST_DWithin, ST_DistanceSphere) filter candidate facilities in <5ms.

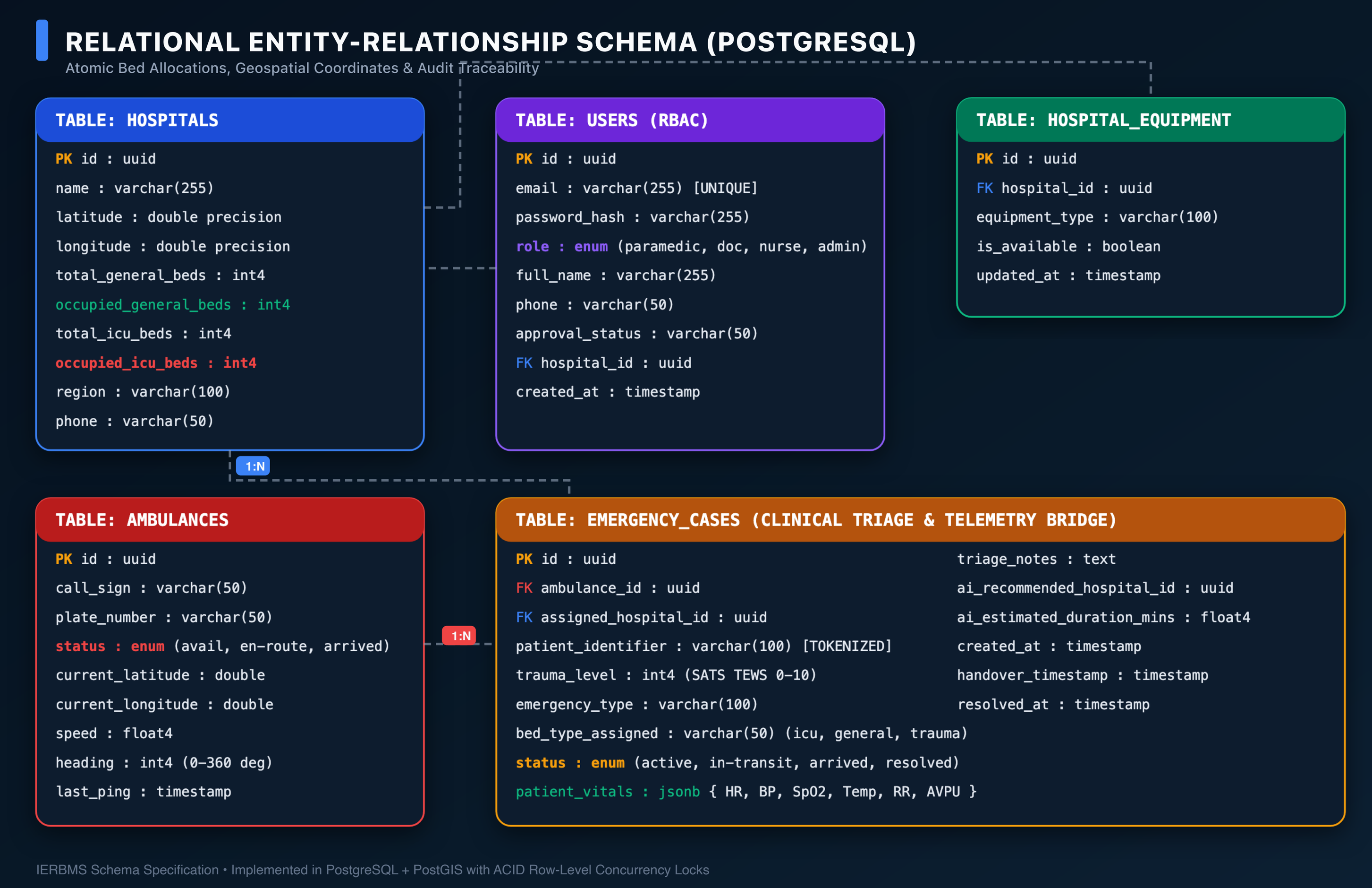

Fig. 2: Relational PostGIS Spatial Schema & Telemetry Data Model

4. FIVE-TIER SYSTEM ARCHITECTURE

DISTRIBUTED MICROSERVICES

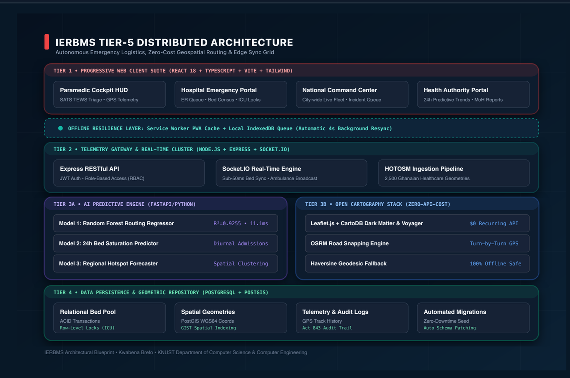

Fig. 3: 5-Tier Pipeline: PWA Edge • Gateway • PostGIS • FastAPI ML • OSRM Engine

5. ZERO-COST OPEN GEOSPATIAL ROUTING

100% FREE & SELF-HOSTED

EVALUATION PARAMETER	COMMERCIAL STACK (GOOGLE MAPS)	PULSEGRID OPEN GEOSPATIAL STACK
Cost per 100k Queries	\$500.00 – \$1,000.00 USD / mo	\$0.00 USD (Self-hosted & Open)
Geographic Dataset	Proprietary Google Network	OpenStreetMap & HOT-OSM Ghana (2,500 facilities)
Routing Machine	Cloud-dependent API	OSRM Road Snapping + Haversine Vector Fallback
Tactical Map Tiles	Raster Standard Google View	CartoDB Dark Matter, Esri Satellite, Street Voyager

6. PARAMEDIC HUD & 3D BED CAPACITY

MISSION-FOCUSED UX

Fig. 4a: Paramedic Cockpit (Turn-by-turn road snapping)

Fig. 4b: Hospital ER 3D Mesh (Pre-arrival bed allocation)

Tactical Mission Isolation: When an ambulance unit accepts a critical emergency, all 2,499 unrelated facility markers vanish from the tactical map. Only the driver's current position, road-snapped OSRM navigation path, and destination hospital are rendered—eliminating driver cognitive overload.

7. MACHINE LEARNING HOSPITAL MATCHING

RANDOM FOREST REGRESSOR

Trained on historical hospital intake patterns. Enforces **Hard Clinical Guardrails** (distance ≤ 60 km, specialized ICU bed > 0 for trauma ≥ 4) + predicts total turnaround latency:

MODEL ARCHITECTURE	R^2 SCORE	RMSE (MINS)	MAE (MINS)	LATENCY
Ordinary Least Squares (OLS)	0.9250	7.14	5.68	1.2 µs
Decision Tree (CART)	0.8672	9.48	7.44	3.5 µs
Multi-Layer Perceptron (DNN)	0.9180	7.42	5.92	142.0 µs
Random Forest (PULSEGRID)	**0.9255**	**7.12**	**5.67**	**11.1 µs**

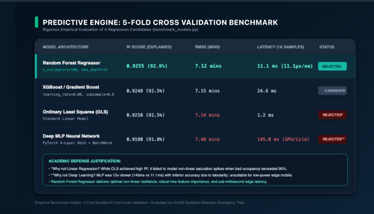

Fig. 5: 5-Fold Cross-Validation: R^2 Fit vs Inference Latency

8. TWO-TIER OFFLINE SYNC ENGINE

DEAD-ZONE FAULT TOLERANT

When transit units enter cellular dead-zones, vitals and triage status updates are intercepted by the **Service Worker** and persisted into an **IndexedDB Local Storage Queue**. The client listens for network recovery (`navigator.onLine`) and flushes the queue via transactional HTTP payloads to PostgreSQL without data loss.

9. QUANTITATIVE CLINICAL IMPACT

SIMULATED BENCHMARK

-35%	+28%	2,500+	\$0.00
AMBULANCE TURNAROUND	ICU UTILIZATION	FACILITIES INGESTED	RECURRING API FEES

10. LIVE DEPLOYMENT & ACT 843

SCAN FOR LIVE DEMO

SCAN WITH PHONE CAMERA
https://final-year-project-frontend-fawn.vercel.app/
Instant interactive test of Paramedic Cockpit, Hospital ER Triage, and National Telemetry Grid across Ghana.

Regulatory Compliance: Formulated in compliance with the **Ghana Data Protection Act, 2012 (Act 843)**. Enforces Role-Based Access Control (RBAC), end-to-end tokenized sessions, and clinical data minimization.

STACK: React 18 · TypeScript · Vite · Leaflet · OSRM · Node.js / Express · PostgreSQL 16 + PostGIS · Python / FastAPI · Random Forest (Scikit-Learn) · Tailwind CSS · IndexedDB PWA